

Elektrodynamik Formelsamling

Mikkel Moth Billing

202307989

25. marts 2025

Indhold

Indhold	1
1 Math	3
1.1 Vectors	3
1.2 Calculus	3
Dirac Delta Function	4
1.3 Fundamental theorem of calculus	4
2 Electrostatics	5
2.1 Electric field	5
2.2 Divergence and Curl	5
2.3 Electric potential	6
2.4 Work and Energy	8
2.5 Conductors	8
Capacitors	10
3 Potential	12
3.1 Laplace's equation	12
3.2 The Method of Images	12
3.3 Separation of variables	13
3.4 Multipole Expansion	14
4 Electric Fields in Matter	15
4.1 Dielectrics	15
Alignment of polar molecules	15
4.2 Field of a Polarized Object	16
4.3 Electric Displacement	17
4.4 Linear Dielectrics	17
5 Magnetostatics	18
5.1 Magnetic Forces	18
Current	19
5.2 Biot-Savart Law	19
5.3 Divergence and Curl of B	20
Ampère's Law	20
5.4 Magnetic Vector Potential	21

6	Magnetic Fields in Matter	22
6.1	Magnetization	22
	Atomic Orbits	23
	Magnetization	24
6.2	Field of a Magnetized Object	24
6.3	Auxiliary Field	24
6.4	Linear and Nonlinear Media	25
	Ferromagnetism	25
7	Electrodynamics	25
7.1	Electromotive Force	25
	Motional emf	26
7.2	Electromagnetic Induction	26
	Inductance	27
7.3	Maxwell's Equations	28
	Maxwell's Equations in Matter	28
8	Conservation Laws	29
8.1	Charge and energy	29
8.2	Momentum	30
9	Appendix	30

1 Math

1.1 Vectors

$$a(\mathbf{A} + \mathbf{B}) = a\mathbf{A} + a\mathbf{B}$$

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos(\theta) \quad (1.1)$$

$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}||\mathbf{B}|\sin(\theta)\hat{\mathbf{n}} \quad (1.4)$$

$$(\mathbf{A} \times \mathbf{B}) = -(\mathbf{B} \times \mathbf{A}) \quad (1.6)$$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \quad (1.15)$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \quad (1.17)$$

$$(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C})$$

$$\mathbf{A} \times [\mathbf{B} \times (\mathbf{C} \times \mathbf{D})] = \mathbf{B}[\mathbf{A} \cdot (\mathbf{C} \times \mathbf{D})] - (\mathbf{A} \cdot \mathbf{B})(\mathbf{C} \cdot \mathbf{D}) \quad (1.18)$$

1.2 Calculus

$$dT = \left(\frac{\partial T}{\partial x} \hat{\mathbf{x}} + \frac{\partial T}{\partial y} \hat{\mathbf{y}} + \frac{\partial T}{\partial z} \hat{\mathbf{z}} \right) \cdot (dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}})$$

$$= (\nabla T) \cdot d\mathbf{l}, \quad (1.35)$$

where the gradient (∇) is an operator:

$$\nabla = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}. \quad (1.39)$$

The gradient is used to calculate the divergence of a vector with:

$$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}, \quad (1.40)$$

and the curl with:

$$\nabla \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix}$$

$$= \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{\mathbf{z}}. \quad (1.41)$$

The product rules for the gradient are:

$$\nabla(fg) = f\nabla g + g\nabla f$$

$$\nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A}$$

$$\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f)$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$$

$$\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) + \nabla f \times \mathbf{A}$$

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) + (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B}$$

When we use the gradient to make a second derivative we get the Laplacian:

$$\nabla(\nabla T) = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}, \quad (1.42)$$

and the laplacian of a vector:

$$\nabla^2 \mathbf{v} = (\nabla^2 v_x) \hat{\mathbf{x}} + (\nabla^2 v_y) \hat{\mathbf{y}} + (\nabla^2 v_z) \hat{\mathbf{z}}. \quad (1.43)$$

We also have the following identities:

$$\nabla \times (\nabla T) = 0 \quad (1.45)$$

$$\nabla \cdot (\nabla \times \mathbf{v}) = 0 \quad (1.46)$$

$$\nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}. \quad (1.47)$$

Dirac Delta Function

$$\begin{aligned} \delta(x-a) &= \begin{cases} 0, & \text{if } x \neq a \\ \infty, & \text{if } x = a \end{cases} \\ \int_{-\infty}^{\infty} \delta(x-a) dx &= 1 \\ \int_{-\infty}^{\infty} f(x) \delta(x-a) dx &= f(a) \\ \int_{-\infty}^{\infty} f(x) \delta(g(x)) dx &= \sum_{x_i} \frac{f(x_i)}{|g'(x_i)|} \\ \delta(x^2 - a^2) &= \frac{\delta(x-a) + \delta(x+a)}{2|a|} \end{aligned}$$

1.3 Fundamental theorem of calculus

Gradient:

$$\int_{\mathbf{a}}^{\mathbf{b}} (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a}) \quad (1.55)$$

Divergence:

$$\int_{\mathcal{V}} (\nabla \cdot \mathbf{v}) dV = \oint_{\mathcal{S}} \mathbf{v} \cdot d\mathbf{a} \quad (1.56)$$

Curl:

$$\int_{\mathcal{S}} (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint_{\mathcal{P}} \mathbf{v} \cdot d\mathbf{l} \quad (1.57)$$

2 Electrostatics

2.1 Electric field

The force on a test charge (Q) from a single point charge (q), a distance $\mathbf{r}$ away, is calculated by Coulomb's law:

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{\mathbf{r}}, \quad (2.1)$$

where $\epsilon_0 = 8.85 \cdot 10^{-12} \frac{\text{C}^2}{\text{N}\cdot\text{m}^2}$ is the permittivity of free space.

If we have several point charges, at different distances from Q , we have an electric field:

$$\mathbf{F} = Q\mathbf{E}, \quad (2.3)$$

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i. \quad (2.4)$$

If we have a continuous charge distribution, we can calculate the electric field by:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r^2} \hat{\mathbf{r}} dq, \quad (2.5)$$

where dq is the charge in a small element, be it line charge ($dq = \lambda dl$), area charge ($dq = \sigma da$) or volume charge ($dq = \rho d\tau$).

Special cases:

Field of infinite length wire:

$$E = \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{z}$$

Field of infinite plane:

$$\mathbf{E} = \frac{\sigma}{2\epsilon_0} \hat{\mathbf{a}}$$

Field outside a uniformly charged sphere with total charge q , $r > R$:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

2.2 Divergence and Curl

Field lines are imaginary lines used to represent the direction and strength of a field, such as an electric field. They start on positive charges and end on negative charges. The density of these lines indicates the field's strength; closer lines mean a stronger field.

Flux refers to the quantity of a field passing through a given surface. For an electric field, it is the product of the electric field strength and the area

through which the field lines pass, considering the angle between the field lines and the surface.

$$\Phi = \int_S \mathbf{E} \cdot d\mathbf{a} \quad (2.11)$$

Gauss's law states that the total electric flux through a closed surface is proportional to the charge enclosed within that surface.

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}, \quad (2.13)$$

where Q_{enc} is the charge enclosed by the surface. It can be rewritten with the divergence theorem, when $Q_{\text{enc}} = \int_V \rho d\tau$:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}. \quad (2.14)$$

To calculate the curl of the electric field we see that:

$$\int_a^b \mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_a} - \frac{q}{r_b} \right), \quad (2.18)$$

which on a closed interval gives us:

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0, \quad (2.19)$$

$$\nabla \times \mathbf{E} = 0. \quad (2.20)$$

2.3 Electric potential

Since we have $\oint \mathbf{E} \cdot d\mathbf{l}$, it must be independent of the path taken, and therefore we can calculate the potential of a point as:

$$V(\mathbf{r}) = - \int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}, \quad (2.21)$$

and the electric potential difference between two points as:

$$V(\mathbf{b}) - V(\mathbf{a}) = - \int_a^b \mathbf{E} \cdot d\mathbf{l}, \quad (2.22)$$

$$\Rightarrow \mathbf{E} = -\nabla V. \quad (2.23)$$

Poisson's equation:

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}, \quad (2.24)$$

which gives us a way to calculate the potential from the charge distribution:

For a collection of point charges:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}, \quad (2.27)$$

For a continuous charge distribution:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{z} dq, \quad (2.28)$$

For a volume charge distribution:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{z} d\tau'. \quad (2.29)$$

Special cases:

Potential of infinite length wire:

$$V = \frac{1}{4\pi\epsilon_0} \lambda \ln \left(\frac{b}{a} \right)$$

Potential of infinite plane:

$$V = \frac{\sigma}{2\epsilon_0} z$$

Potential of spherical shell:

$$V = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{q}{r} & r > R \\ \frac{1}{4\pi\epsilon_0} \frac{q}{R} & r < R \end{cases}$$

Figure (2.35) shows the way to go between electric field, electric potential and volume charge:

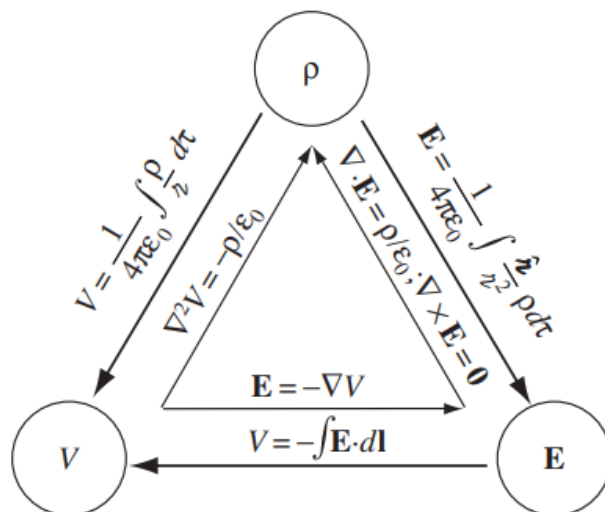


FIGURE 2.35

2.4 Work and Energy

If you want to move a test charge (Q) from point $\mathbf{a}$ to point $\mathbf{b}$, you have to do work against the electric field. The work done is the potential difference between the two points:

$$W = Q [V(\mathbf{b}) - V(\mathbf{a})]. \quad (2.38)$$

So if you want to move the point from infinity to a point $\mathbf{r}$, and you have the reference point in infinity:

$$W = QV(\mathbf{r}). \quad (2.39)$$

The work to move a collection of charges is:

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \sum_{j>i}^n \frac{q_i q_j}{r_{ij}}, \quad (2.40)$$

$$= \frac{1}{8\pi\epsilon_0} \sum_{i=1}^n \sum_{j \neq i}^n \frac{q_i q_j}{r_{ij}}, \quad (2.41)$$

$$= \frac{1}{2} \sum_{i=1}^n q_i V(\mathbf{r}_i). \quad (2.42)$$

For continuous charge distributions:

$$W = \frac{1}{2} \int V dq, \quad (2.43)$$

where again, dq is the charge in a small element, be it line charge ($dq = \lambda dl$), area charge ($dq = \sigma da$) or volume charge ($dq = \rho d\tau$).

If we take (2.43) for volume over all space, we get:

$$W = \frac{\epsilon_0}{2} \int E^2 d\tau. \quad (2.45)$$

Special cases

Energy of Q spread uniformly throughout sphere of radius R :

$$\begin{aligned} W_{tot} &= W_{out} + W_{in} \\ &= \frac{Q^2}{8\pi\epsilon_0 R} + \frac{3Q^2}{20\pi\epsilon_0 R} \end{aligned}$$

2.5 Conductors

A conductor is a material where the electrons are free to move around. The electric field inside a conductor is zero, and the charges are distributed on the surface, like metal. The electric field inside a conductor is zero, and when it

is placed in an electric field, the charges collect on the surface to create an electrical field inside the conductor, that cancels out the initial field.

$$\nabla \mathbf{E} = \frac{\rho}{\epsilon_0} = 0$$

Induced charges on a conductor are always on the surface, and the electric field is always perpendicular to the surface. For a conductor with a cavity, the electric field is zero inside the cavity, and the charges are on the surface of the cavity. Outside the conductor, the electric field is the same as if the conductor was not there.

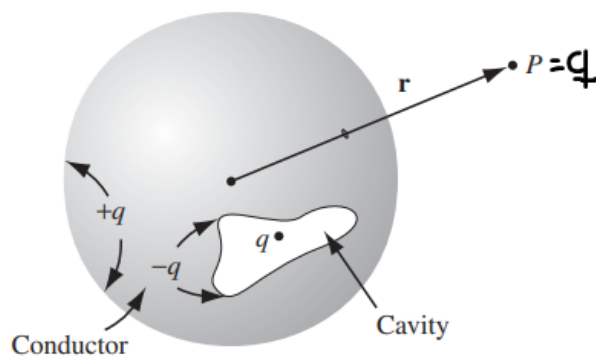


FIGURE 2.46

Figure 3.24 shows a conductor in an electric field, and how it would bend the field lines, so they are always perpendicular. so the electric field parallel to a conductor is 0.

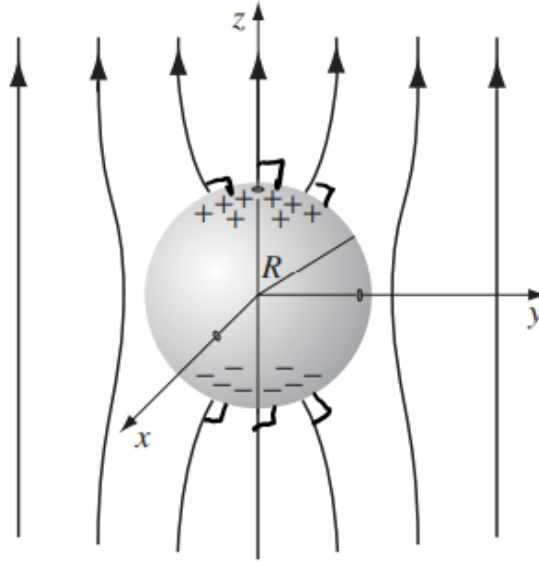


FIGURE 3.24

Because the field inside a conductor is zero, the field immediately outside is:

$$\mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}, \quad (2.48)$$

The force per unit area and outward electrostatic pressure on the surface of a conductor is:

$$\mathbf{f} = \frac{1}{2\epsilon_0} \sigma^2 \hat{\mathbf{n}}, \quad (2.51)$$

$$P = \frac{\epsilon_0}{2} E^2 = \frac{\sigma^2}{2\epsilon_0}. \quad (2.52)$$

Capacitors

If we have two conductors with charge Q and $-Q$, the electric potential between them is:

$$V = V_+ - V_- = - \int_{(-)}^{(+)} \mathbf{E} \cdot d\mathbf{l},$$

if the shape is complicated, $\mathbf{E}$ is hard to calculate, but we know it is proportional to Q , and therefore V is also proportional, which gives us the capacitance:

$$C = \frac{Q}{V}. \quad (2.53)$$

We can talk about the capacitance of a single capacitor, this refers to (2.53) where the reference point for V is infinity.

For a parallel-plate capacitor the capacitance is:

$$C = \frac{A\epsilon_0}{d}, \quad (2.54)$$

To charge a capacitor you have to move the charge from the positive plate to the negative plate. the energy required to do this is:

$$W = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C}. \quad (2.55)$$

3 Potential

3.1 Laplace's equation

To easily calculate the potential in a region, we can use Laplace's equation:

$$\nabla^2 V = 0. \quad (3.4)$$

In one dimension the solution is:

$$V(x) = ax + b, \quad (3.6)$$

but for 2 dimensions it is not easy to calculate exactly, but you can calculate $V(x, y)$ as the average of the potential around the point:

$$V(x, y) = \frac{1}{2\pi R} \oint_{circle} V dl,$$

which might make no sense, but it can be used to calculate the charge distribution, when boundary conditions are known.

In three dimensions the value of V at point $\mathbf{r}$ is the average value of V over a spherical surface of radius R centered at $\mathbf{r}$:

$$V(\mathbf{r}) = \frac{1}{4\pi R^2} \oint_{sphere} V da.$$

First uniqueness theorem: The solution to Laplace's equation in some volume $\mathcal{V}$ is uniquely determined if V is specified on the boundary surface $\mathcal{S}$. Meaning that it doesn't matter how you get your solution, if it satisfies Laplace's equation, and the boundary values are correct, then the answer is correct.

Second uniqueness theorem: In a volume $\mathcal{V}$ surrounded by conductors and containing a specified charge density ρ , the electric field is uniquely determined if the *total charge* on each conductor is given. (The region as a whole can be bounded by another conductor, or else unbounded.)

3.2 The Method of Images

The method of images can be used to solve advanced electrostatic problems, by replacing conductors with fictitious charges that mimic the same boundary conditions.

For example, we have a infinite grounded plate, with a point charge a distance d above it. The boundary potential is that $V = 0$ when $z = 0$ and that $V \rightarrow 0$ far from the charge (Fig 3.10). The same boundary conditions are satisfied if we have a point charge at $-d$ instead of the conductor (Fig 3.11). Therefore the potential for a pointcharge above an infinite grounded plate is:

$$V(x, y, z) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z + d)^2}} \right]. \quad (3.9)$$

From this we can calculate the induced surface charge on the conductor:

$$\begin{aligned}\sigma(x, y) &= -\epsilon_0 \frac{\partial V}{\partial z} \Big|_{z=0} \\ &= \frac{-qd}{2\pi(x^2 + y^2 + d^2)^{3/2}},\end{aligned}\tag{3.10}$$

and the total charge:

$$Q = \int \sigma da = -q.\tag{3.11}$$

We can also use the method of images to find the force on the point charge due to the induction:

$$\mathbf{F} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \hat{\mathbf{z}},\tag{3.12}$$

but the work will not be the same for the plane, and the point charge, so if we want to calculate the energy to move the charge from infinity to d as:

$$W = \int_{\infty}^d \mathbf{F} \cdot d\mathbf{l} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d}.\tag{3.14}$$

You can not put the image charge in the region where you are calculating the potential, so in example 3.2, where we have a point charge a distance a away from a spherical grounded conductor, the image charge is inside the sphere.

This gives the potential:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\tau} + \frac{q'}{\tau'} \right),\tag{3.17}$$

where q' is the image charge, and τ and τ' is the distance from q and q' .

3.3 Seperation of variables

Chapter 3.3.1 shows and 3 examples of how to calculate the potential between 2 parallel grounded conductors, connected by insulators. To calculate the potential between the plates we use separation of variables, so we assume the potential can be written as:

$$V(x, y) = X(x)Y(y)\tag{3.22}$$

$$\Rightarrow \frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = 0\tag{3.23}$$

which we can then solve from the boundary conditions of the problem, with the help of Fourier's trick.

In chapter 3.3.2 we find a general solution for the potential in spherical coordinates, with symmetry in φ :

$$V(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos(\theta)),\tag{3.65}$$

where A_l and B_l are constants determined by the boundary conditions, and $P_l(\cos(\theta))$ is the Legendre polynomials given by:

$$P_l(x) = \frac{1}{2^l l!} \left(\frac{d}{dx} \right)^l (x^2 - 1)^l. \quad (3.62)$$

$P_0(x) = 1$
$P_1(x) = x$
$P_2(x) = (3x^2 - 1)/2$
$P_3(x) = (5x^3 - 3x)/2$
$P_4(x) = (35x^4 - 30x^2 + 3)/8$
$P_5(x) = (63x^5 - 70x^3 + 15x)/8$

TABLE 3.1 Legendre Polynomials.

3.4 Multipole Expansion

A dipole is two opposite charges particles, collect two dipoles to make an quadropole, and so on.

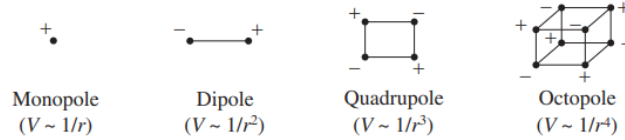


Fig. 3.27

To calculate the potential from a multipole we use:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n P_n(\cos(\alpha)) \rho(\mathbf{r}') d\tau', \quad (3.95)$$

where P_l is the Legendre polynomial, ρ is the total charge of the distribution. $n = 0$ for monopole, 1 for dipole and so on.

For monopole and dipole we get:

$$V_{mon}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}, \quad (3.97)$$

$$V_{dip}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2}, \quad (3.99)$$

where $\mathbf{p}$ is the dipole moment of the distribution:

$$\mathbf{p} = \int \mathbf{r}' \rho(\mathbf{r}') d\tau'. \quad (3.98)$$

The electric field of a dipole is:

$$\mathbf{E}_{dip}(r, \theta) = \frac{p}{4\pi\epsilon_0 r^3} (2 \cos(\theta) \hat{\mathbf{r}} + \sin(\theta) \hat{\boldsymbol{\theta}}). \quad (3.103)$$

4 Electric Fields in Matter

4.1 Dielectrics

If you put a neutral atom in an electric field, it is either ripped apart, being ionized, or it reaches an equilibrium, where the force from the electric field and the force holding the atom together cancel each other, making the atom polarized. This will cause the atom to have a dipole moment $\mathbf{p}$, with the same direction as $\mathbf{E}$.

$$\mathbf{p} = \alpha \mathbf{E}, \quad (4.1)$$

here α is called the atomic polarizability, and is dependent on the atom's structure.

If an atom was made by a center charge q surrounded by a spherical charged cloud $-q$ with radius a , then the electric field we place the atom in, would move the nucleus a distance d from the center. Now the electric field in equilibrium would be:

$$\begin{aligned} E &= \frac{1}{4\pi\epsilon_0} \frac{qd}{a^3}, \quad \text{or} \quad p = qd = (4\pi\epsilon_0 a^3) E, \\ \Rightarrow \alpha &= 4\pi\epsilon_0 a^3 = 3\epsilon_0 v, \end{aligned} \quad (4.2)$$

where v is the volume of the atom.

Table 4.1. Atomic polarizabilities ($\alpha/4\pi\epsilon_0$, in units of 10^{-30} m^3). Data from <i>Handbook of Chemistry and Physics</i> , 91st ed., CRC Press, Boca Raton (2010).									
H	He	Li	Be	C	Ne	Na	Ar	K	Cs
0.667	0.205	24.3	5.60	1.67	0.396	24.1	1.64	43.4	59.4

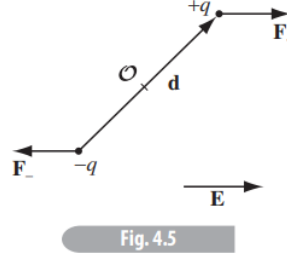
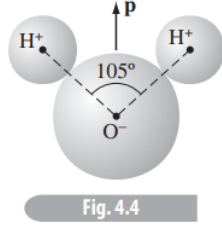
For molecules it is not so easy, as the dipole moment might not be the same direction as $\mathbf{E}$, and therefore we get 9 polarizability constants:

$$\begin{cases} p_x = \alpha_{xx} E_x + \alpha_{xy} E_y + \alpha_{xz} E_z, \\ p_y = \alpha_{yx} E_x + \alpha_{yy} E_y + \alpha_{yz} E_z, \\ p_z = \alpha_{zx} E_x + \alpha_{zy} E_y + \alpha_{zz} E_z. \end{cases} \quad (4.3)$$

Alignment of polar molecules

Some molecules have their own dipole moment, such as water, which are called polar molecules. If placed in a uniform electric field, the force on the positive end, cancels out with the force on the negative end, but there will be a torque

$$\mathbf{N} = (\mathbf{r}_+ \times \mathbf{F}_+) + (\mathbf{r}_- \times \mathbf{F}_-) = \mathbf{p} \times \mathbf{E}. \quad (4.4)$$



$\mathbf{N}$ is in such a direction to line $\mathbf{p}$ parallel to $\mathbf{E}$.

If the field is not uniform over det atom, which is rare, there is a net force on the atom

$$\mathbf{F} = (\mathbf{p} \cdot \nabla) \mathbf{E}. \quad (4.5)$$

An ideal dipole is the atrifical limit $d \rightarrow 0$ and $q \rightarrow \infty$. The energy of an ideal dipolemoment is

$$U = -\mathbf{p} \cdot \mathbf{E}, \quad (4.6)$$

and the interaction energy between two ideal dipoles with displacement $\mathbf{r}$ is

$$U = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [\mathbf{p}_1 \cdot \mathbf{p}_2 - 3(\mathbf{p}_1 \cdot \hat{\mathbf{r}})(\mathbf{p}_2 \cdot \hat{\mathbf{r}})]. \quad (4.7)$$

If we place a material made of nonpolar molecules in an electric field, it will induce a dipole moment in all the molecules. If we place a polar material, the field will turn all the molecules. We get the polarization which is

$\mathbf{P}$ = dipole moment per unit volume.

4.2 Field of a Polarized Object

To calculate the field that arises from the induced dipoles in a material placed in an electric field, we first calculate the potential by integrating each dipole over det volume $\mathbf{p} = \mathbf{P}d\tau'$:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\mathbf{P}(\mathbf{r}') \cdot \hat{\mathbf{z}}}{z^2} d\tau', \quad (4.9)$$

$$= \frac{1}{4\pi\epsilon_0} \oint_S \frac{\sigma_b}{z} da' + \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho_b}{z} d\tau', \quad (4.13)$$

where we have the surface charge density $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ and the volume charge density $\rho_b = -\nabla \cdot \mathbf{P}$.

For a uniformly polarized sphere of radius R , the volume charge density is zero baccuse $\mathbf{P}$ is uniform, but

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = P \cos \theta,$$

from this we get

$$V(r, \theta) = \begin{cases} \frac{P}{3\epsilon_0} r \cos \theta & \text{for } r \leq R, \\ \frac{P}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta & \text{for } r \geq R. \end{cases}$$

So the field inside the sphere is uniform, and outside the sphere it behaves like a ideal dipole, whose dipole moment is the total dipole moment of a sphere $\mathbf{p} = \frac{4}{3}\pi R^3 \mathbf{P}$.

4.3 Electric Displacement

Along with the bound charge, there is also free charge which could be electrons on a conductor, or ions embedded in the dielectric material, so any charge not associated with $\mathbf{P}$. Within the dielectric, the total charge density is

$$\rho = \rho_b + \rho_f, \quad (4.20)$$

and Gauss's law is

$$\epsilon_0 \nabla \cdot \mathbf{E} = \rho = \rho_b + \rho_f = -\nabla \cdot \mathbf{P} + \rho_f,$$

where $\mathbf{E}$ is the total field.

This leads to a term for ρ_f :

$$\begin{aligned} \rho_f &= \nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}), \\ &= \nabla \cdot \mathbf{D}, \end{aligned} \quad (4.22)$$

$$\Rightarrow \oint \mathbf{D} \cdot d\mathbf{a} = Q_{f_{enc}}. \quad (4.23)$$

Here $\mathbf{D}$ is known as the electric displacement.

Typically we know ρ_f , and we can therefore calculate $\mathbf{D}$. When computing the displacement look for symmetry, if there is spherical, cylindrical or plane symmetry we can use (4.23) because $\nabla \times \mathbf{D} = 0$.

4.4 Linear Dielectrics

The polarization is proportional to the electric field:

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}, \quad (4.30)$$

where χ_e is the electric susceptibility, which is dependant on the structure of the material. Materials that obey eq. 4.30 are called linear dielectrics. Notice that $\mathbf{E}$ is the total electric field, and therefore we can not calculate $\mathbf{P}$ from eq. 4.30, instead we could calculate

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} = \epsilon_0 (1 + \chi_e) \mathbf{E} = \epsilon \mathbf{E}, \quad (4.31)$$

where we have the permittivity of the material, and dielectric constant (Table 4.2):

$$\epsilon = \epsilon_0 (1 + \chi_e), \quad (4.33)$$

$$\epsilon_r = 1 + \chi_e = \frac{\epsilon}{\epsilon_0}. \quad (4.34)$$

When we have a charge distribution in homogeneous linear dielectric, the field is calculated as in vacuum, but using ϵ instead of ϵ_0 . To calculate the full

energy stored in a electrostatic system in a linear dielectric, we rewrite (2.45) to

$$W = \frac{\epsilon_0}{2} \int \epsilon_r E^2 d\tau = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau. \quad (4.58)$$

Say we have a dielectric in a capacitor, the force required to remove the dielectric can be calculated as

$$F = -\frac{dW}{dx} = \frac{1}{2} V^2 \frac{dC}{dx}, \quad (4.64)$$

$$= -\frac{\epsilon_0 \chi_e w}{2d} V^2, \quad (4.65)$$

where d is the distance between the plates, and w is the width of the plate, perpendicular to the force.

5 Magnetostatics

5.1 Magnetic Forces

If we have a wire with a current running through, it will create a magnetic field, the direction of the field can be found by the right hand rule, if your thumb is in the direction of current, your fingers curl in the direction of the magnetic field.

A force will be induced on a moving charge (Q) in a magnetic field, moving with velocity ($\mathbf{v}$)

$$\mathbf{F}_{mag} = Q (\mathbf{v} \times \mathbf{B}). \quad (5.1)$$

This is the Lorentz force law. When there is both a magnetic field and an electric field, the force would be

$$\mathbf{F} = Q (\mathbf{E} + (\mathbf{v} \times \mathbf{B})). \quad (5.2)$$

For a charged particle in a magnetic field, the magnetic force will create a centripetal acceleration, which leads to the cyclotron formula, used to find the momentum of a charged particle,

$$QvB = m \frac{v^2}{R}, \quad \text{or } p = QBR. \quad (5.3)$$

If we have both a magnetic field and an electric field, we can follow example 5.2, to show that the particle will move in a cycloid motion, a circle where the center moves $(0, R\omega t, R)$:

$$(y - R\omega t)^2 + (z - R)^2 = R^2 \quad (5.9)$$

$$R\omega = \frac{E}{B}. \quad (5.10)$$

A magnetic field can not speed up or slow down a particle, it can only change its direction, therefore the field does no work on the particle,

$$dW_{mag} = \mathbf{F}_{mag} \cdot d\mathbf{l} = Q(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} dt = 0. \quad (5.11)$$

Current

A line charge λ traveling down a wire at speed v , constitutes a charge

$$\mathbf{I} = \lambda \mathbf{v}. \quad (5.14)$$

The magnetic force on a segment of current-carrying wire is:

$$\mathbf{F}_{mag} = \int (\mathbf{I} \times \mathbf{B}) d\mathbf{l}, \quad (5.15)$$

if $\mathbf{I}$ and $d\mathbf{l}$ point in the same direction:

$$\mathbf{F}_{mag} = I \int (d\mathbf{l} \times \mathbf{B}). \quad (5.17)$$

When charge moves over a surface we describe it by the surface current density

$$\mathbf{K} = \frac{d\mathbf{I}}{dl_{\perp}} = \sigma \mathbf{v}. \quad (5.23)$$

And the magnetic force on the surface current is

$$\mathbf{F}_{mag} = \int (\mathbf{K} \times \mathbf{B}) d\mathbf{a}. \quad (5.24)$$

In a volume we have the volume current density:

$$\mathbf{J} = \frac{d\mathbf{I}}{da_{\perp}} = \rho \mathbf{v}, \quad (5.26)$$

$$\mathbf{F}_{mag} = \int (\mathbf{J} \times \mathbf{B}) d\tau. \quad (5.27)$$

The total charge crossing a surface can be calculated as

$$I = \int_S \mathbf{J} d\mathbf{a}_{\perp} = \int_S \mathbf{J} d\mathbf{a}. \quad (5.28)$$

From this we get the continuity equation

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}. \quad (5.29)$$

5.2 Biot-Savart Law

The magnetic field of a steady line current is given by Biot-Savart law:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I} \times \hat{\mathbf{r}}}{r^2} dl',$$

where μ_0 is the permeability of free space.

The magnitude of the magnetic field caused by an infinite wire is:

$$B = \frac{\mu_0 I}{2\pi s}, \quad (5.38)$$

where the direction follows the right-hand rule. For a point a distance z above a circular loop of radius R , the field is:

$$B(z) = \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + z^2)^{3/2}}. \quad (5.41)$$

For surface and volume charge the Biot-Savart law becomes:

$$\begin{aligned} \mathbf{B}(\mathbf{r}) &= \frac{\mu_0}{4\pi} \int \frac{\mathbf{K} \times \hat{\mathbf{r}}}{r^2} da', \\ \mathbf{B}(\mathbf{r}) &= \frac{\mu_0}{4\pi} \int \frac{\mathbf{J} \times \hat{\mathbf{r}}}{r^2} d\tau'. \end{aligned} \quad (5.42)$$

5.3 Divergence and Curl of $\mathbf{B}$

For any wire running through a chosen loop, the integral of $\mathbf{B}$ is:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}, \quad (5.44)$$

where I_{enc} is the total current enclosed by the loop. If the flow of current is volume current density, the enclosed current is:

$$I_{\text{enc}} = \int \mathbf{J} \cdot d\mathbf{a}. \quad (5.45)$$

Applying Stokes law, gives us the curl of $\mathbf{B}$ for infinite wires:

$$\begin{aligned} \int (\nabla \times \mathbf{B}) \cdot d\mathbf{a} &= \mu_0 \int \mathbf{J} \cdot d\mathbf{a}, \\ \Rightarrow \nabla \times \mathbf{B} &= \mu_0 \mathbf{J}. \end{aligned} \quad (5.46)$$

To find the general curl of $\mathbf{B}$, we use Biot Savart law:

$$\nabla \times \mathbf{B} = \frac{\mu_0}{4\pi} \int \mathbf{J} \times \frac{\hat{\mathbf{r}}}{r^2} d\tau', \quad (5.51)$$

$$= \mu_0 \mathbf{J}. \quad (5.56)$$

For the divergence of $\mathbf{B}$ we get:

$$\nabla \cdot \mathbf{B} = 0. \quad (5.50)$$

Ampère's Law

Equation (5.56) is called Ampère's law in differential form, if we integrate we get:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int \mathbf{J} \cdot d\mathbf{a} = \mu_0 I_{\text{enc}}. \quad (5.57)$$

This plays the role of Gauss law for magnetostatics.

Special cases:

Field of infinite sheet with in xy-plane with $\mathbf{K} = K\hat{\mathbf{x}}$:

$$\mathbf{B} = \begin{cases} +\frac{\mu_0}{2}K\hat{\mathbf{y}} & \text{for } z < 0, \\ -\frac{\mu_0}{2}K\hat{\mathbf{y}} & \text{for } z > 0. \end{cases} \quad (5.58)$$

Field of long solenoid with n windings and radius R :

$$\mathbf{B} = \begin{cases} \mu_0 n I \hat{\mathbf{z}} & \text{for } r < R, \\ 0 & \text{for } r > R. \end{cases} \quad (5.59)$$

Field of a toroidal coil (Fig 5.37) with n turns a distance s form the symmetry axis:

$$\mathbf{B}(\mathbf{r}) = \begin{cases} \frac{\mu_0 n I}{2\pi s} \hat{\boldsymbol{\phi}} & \text{inside coil,} \\ 0 & \text{outside coil.} \end{cases} \quad (5.60)$$

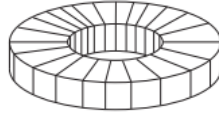


Fig. 5.37

Crosssection can be any shape

5.4 Magnetic Vector Potential

We introduce the vector potential $\mathbf{A}$:

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad (5.61)$$

which divergence is 0:

$$\nabla \cdot \mathbf{A} = 0. \quad (5.63)$$

With the help of Amperes law this extends to:

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}, \quad (5.64)$$

which gives us a direct term for $\mathbf{A}$, if $\mathbf{J}$ goes to zero at infinity:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{\tau} d\tau'. \quad (5.65)$$

If $\mathbf{J}$ doesn't go to zero at infinity we can use:

$$\oint \mathbf{A} \cdot d\mathbf{l} = \int \mathbf{B} \cdot d\mathbf{a} = \Phi. \quad (5.71)$$

Figure 5.51 shows the calculations between $\mathbf{J}$, $\mathbf{B}$ and $\mathbf{A}$.

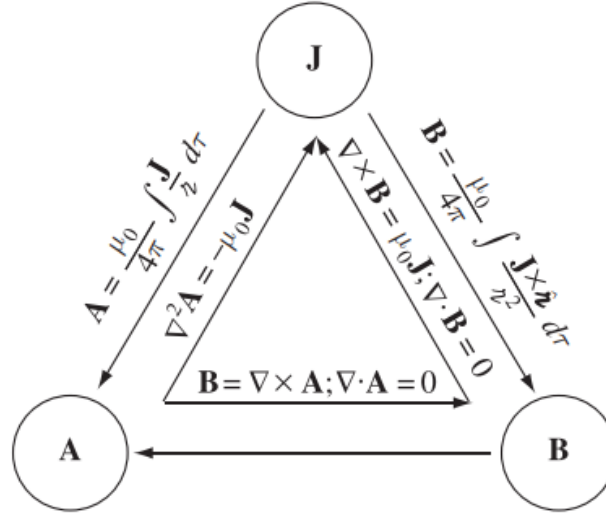


Fig. 5.51

If you want to find the vector potential for a current distribution, we can use multipole expansion:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \oint (r')^n P_n(\cos \alpha) d\mathbf{l}'. \quad (5.80)$$

which in most cases leads to the dipole moment being the dominant term:

$$\mathbf{A}_{\text{dip}}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \hat{\mathbf{r}}}{r^2}, \quad (5.85)$$

where $\mathbf{m}$ is the magnetic dipole moment:

$$\mathbf{m} = I \int d\mathbf{a} = I\mathbf{a}. \quad (5.86)$$

6 Magnetic Fields in Matter

6.1 Magnetization

Magnetic material, are magnetic because the electrons spin around their core. this creates magnetic dipoles. which in most magnets are turned in the direction of the magnetic field, the material is placed in.

If we put a rectangular loop in a uniform field, the loop experiences a torque:

$$\mathbf{N} = aF \sin \theta \hat{\mathbf{x}},$$

the magnitude of the force is $F = IbB$, which leads to:

$$\mathbf{N} = \mathbf{m} \times \mathbf{B}, \quad (6.1)$$

where $\mathbf{m} = Iab$ is the magnetic dipole moment of the loop.

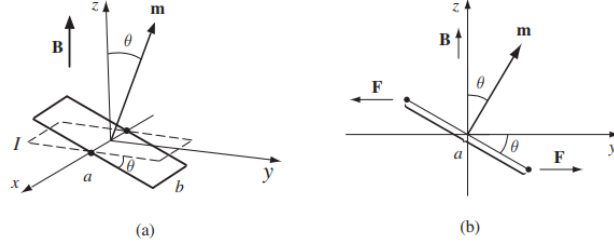


Fig. 6.2

In a uniform magnetic field the net force on a current loop is 0, but in a nonuniform field it is:

$$\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B}). \quad (6.3)$$

Atomic Orbits

If we say electrons revolve around the nucleus in a circular motion with radius R , it will look like a steady current:

$$I = \frac{e}{T} = \frac{e}{2\pi R/v},$$

$$\Rightarrow \mathbf{m} = -\frac{1}{2}evR\hat{\mathbf{z}}. \quad (6.4)$$

If we put an atom in a magnetic field, it will be hard to tilt the orbit of the electron, however the electron will speed up or slow down, depending on the direction of $\mathbf{B}$. Without the field we have:

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{R^2} = m_e \frac{v^2}{R}, \quad (6.5)$$

but in a field there is a force $-e(\mathbf{v} \times \mathbf{B})$, if $\mathbf{B}$ is perpendicular on the plane of orbit, we get:

$$\frac{1}{4\pi\epsilon_0} \frac{e^2}{R^2} + e\bar{v}B = m_e \frac{\bar{v}^2}{R}, \quad (6.6)$$

where $\bar{v}$ is the new speed of the electron. We can calculate the difference assuming the change in speed is small:

$$\Delta v = \bar{v} - v = \frac{eRB}{2m_e}. \quad (6.7)$$

This also leads to a change in dipole moment, opposite the fields direction:

$$\Delta \mathbf{m} = -\frac{1}{2}eR\Delta v\hat{\mathbf{z}} = -\frac{e^2R^2}{4m_e}\mathbf{B}. \quad (6.8)$$

Magnetization

So far we have seen two kinds of Magnetization. *Paramagnetism* where the dipoles associated with spins of unpaired electrons experience a torque tending to line them up parallel to the field, and *Diamagnetism* where the speed of the electrons is altered in such a way to change the orbital dipole moment in a direction opposite the field. These effects can be described by the *magnetization* $\mathbf{M}$, which is the magnetic dipole moment per unit volume.

6.2 Field of a Magnetized Object

With the magnetization we can calculate the vector potential from (5.85) knowing that $\mathbf{m} = \mathbf{M}d\tau'$:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{M}(\mathbf{r}') \times \hat{\mathbf{z}}}{r^2} d\tau', \quad (6.11)$$

which can also be rewritten in terms of the potential of a volume current $\mathbf{J}_b = \nabla \times \mathbf{M}$ and surface current $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}_b(\mathbf{r}')}{r} d\tau' + \frac{\mu_0}{4\pi} \oint_S \frac{\mathbf{K}_b(\mathbf{r}')}{r} da'. \quad (6.15)$$

6.3 Auxiliary Field

We can now collect the field due to bound charge, and the field due to the free charge.

$$\mathbf{J} = \mathbf{J}_b + \mathbf{J}_f. \quad (6.17)$$

With Ampere's law we get:

$$\begin{aligned} \frac{1}{\mu_0} (\nabla \times \mathbf{B}) &= \mathbf{J} = \mathbf{J}_b + \mathbf{J}_f = \mathbf{J}_f + (\nabla \times \mathbf{M}), \\ \Rightarrow \nabla \times \left(\frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \right) &= \mathbf{J}_f, \end{aligned}$$

We define $\mathbf{H}$:

$$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}, \quad (6.18)$$

$$\Rightarrow \nabla \times \mathbf{H} = \mathbf{J}_f, \quad (6.19)$$

and in integral form:

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{fenc}}, \quad (6.20)$$

where I_{fenc} is the free current through the loop.

6.4 Linear and Nonlinear Media

In paramagnetic and diamagnetic materials, the magnetization is sustained by the field. Which means $\mathbf{M}$ is proportional to $\mathbf{B}$ and therefore also $\mathbf{H}$:

$$\mathbf{M} = \chi_m \mathbf{H}, \quad (6.29)$$

where χ_m is the magnetic susceptibility, and is dependent of material. Materials that obey (6.29) are called linear media. This gives us:

$$\mathbf{B} = \mu \mathbf{H}, \quad (6.31)$$

$$\Rightarrow \mu = \mu_0(1 + \chi_m), \quad (6.32)$$

which is the permeability of the material (Table 6.1).

Ferromagnetism

In ferromagnetic materials, the alignment of dipoles is "frozen", because each dipole like to face the same way as their neighbours. We split the material up into domains, which can have different alignments, which explains why not all iron tools are magnetic. When a magnetic field is applied, the domains will compete and move their boundaries, so the domains with dipoles facing parallel to the field will grow.

7 Electrodynamics

7.1 Electromotive Force

To make current flow, you have to "push" the charges. For most materials the current density $\mathbf{J}$ is proportional to the force per unit charge $\mathbf{f}$:

$$\mathbf{J} = \sigma \mathbf{f}, \quad (7.1)$$

where σ is the conductivity of the material, which is sometimes written as resistivity $\rho = 1/\sigma$ (Table 7.1). Usually it is the electromagnetic forces that "pushes" and (7.1) becomes:

$$\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad (7.2)$$

but the velocity of the charges is small enough that we can ignore the term, and we get Ohm's Law:

$$\mathbf{J} = \sigma \mathbf{E}, \quad (7.3)$$

which might not seem familiar, but if we use (7.3) to find the potential difference between two electrodes, we get:

$$V = IR. \quad (7.4)$$

As electrons travel down a wire, they collide with the atoms in the material, and each other. This results in energy given off as heat in the resistor:

$$P = VI = I^2 R. \quad (7.7)$$

The reason a light turns on instantly when you close a switch, is because if the charges built up at a point in a circuit, it will create an electric field, that will then restore uniform flow. This electric field propagates through the circuit at lightspeed, hence why the light turns on almost instantly.

So there are two forces that drive the current, the electric field and the source, often chemical force from a battery:

$$\mathbf{f} = \mathbf{f}_s + \mathbf{E}. \quad (7.8)$$

The electromotive force or *emf*, is determined by:

$$\mathcal{E} = \oint \mathbf{f} \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l}. \quad (7.9)$$

In an ideal emf we have $\mathbf{E} = -\mathbf{f}_s$, and therefore:

$$V = \int_a^b \mathbf{f}_s \cdot d\mathbf{l} = \mathcal{E}. \quad (7.10)$$

Motional emf

Suppose you move part of a current loop through a uniform magnetic field, with Lorentz law this will induce a current in the loop.

$$\mathcal{E} = \oint \mathbf{f}_{\text{mag}} \cdot d\mathbf{l} = vBh, \quad (7.11)$$

where the loop is rectangular with height h . But the magnetic force does not do work on the loop, so the person moving the loop is doing the work. The force required is $f_{\text{pull}} = uB$ where u is the vertical speed of the charges. This leads to the work done by unit charge:

$$\oint \mathbf{f}_{\text{pull}} \cdot d\mathbf{l} = vBh = \mathcal{E}.$$

Instead of calculating for each charge, we use the flux through the loop:

$$\begin{aligned} \Phi &= xBh, & \frac{d\Phi}{dt} &= -vBh, \\ \Rightarrow \mathcal{E} &= -\frac{d\Phi}{dt}. \end{aligned} \quad (7.13)$$

7.2 Electromagnetic Induction

Faraday found that if he moved the magnetic field instead of the loop, current would still run through the loop. But it can't be the magnetic field moving the charges, instead it is an induced electric field.

$$\mathcal{E} = \oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt}, \quad (7.23)$$

this leads to Faraday's law:

$$\oint \mathbf{E} \cdot d\mathbf{l} = - \int \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a} \quad (7.24)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (7.25)$$

Toghether with this Lenz's law say that the induced current is in such a direction that it opposes the change in magnetic flux.

If $\mathbf{E}$ is a pure Faraday field, due only to the change in $\mathbf{B}$, then the calculations are analogue to magnetostatics:

$$\mathbf{E} = -\frac{1}{4\pi} \frac{\partial}{\partial t} \int \frac{\mathbf{B} \times \hat{\mathbf{r}}}{r^2} d\tau, \quad (7.27)$$

and if we have symmetry we can use Amperes law:

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt}. \quad (7.28)$$

Inductance

If we have two loops of wire, and a current runs in one of the wires, this creates a magnetic field. This magnetic field will have a flux through the other loop, which is proportional to the current:

$$\Phi = MI, \quad (7.31)$$

where M is the mutual inductance. M is a geometrical property, and is the same for if the current was running in loop 2.

$$M_{12} = M_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{d\mathbf{l}_2 \cdot d\mathbf{l}_1}{r}. \quad (7.32)$$

If the current is not steady we see that an emf is induced in the other loop:

$$\mathcal{E}_2 = -\frac{d\Phi_2}{dt} = -M \frac{dI_1}{dt}. \quad (7.34)$$

On top of that it also induces an emf in itself:

$$\Phi = LI, \quad (7.35)$$

where L is the self-inductance, or just inductance. If the current is changing we get:

$$\mathcal{E} = -L \frac{dI}{dt}. \quad (7.36)$$

When you close a circuit with an inductance L and resistance R , the current will build up, since

$$\begin{aligned} \mathcal{E}_0 - L \frac{dI}{dt} &= IR, \\ \Rightarrow I(t) &= \frac{\mathcal{E}_0}{R} (1 - e^{-(R/L)t}), \end{aligned} \quad (7.38)$$

if $I(0) = 0$.

It takes energy to build up this current in the loop, which can be seen as energy stored in the magnetic field. If we start at $I(0) = 0$ and work up to I ,

we get:

$$W = \frac{1}{2}LI^2, \quad (7.39)$$

$$= \frac{1}{2} \oint (\mathbf{A} \cdot \mathbf{I}) dl, \quad (7.40)$$

$$= \frac{1}{2} \int_{\mathcal{V}} (\mathbf{A} \cdot \mathbf{J}) d\tau, \quad (7.41)$$

$$= \frac{1}{2\mu_0} \int_{\text{all space}} B^2 d\tau. \quad (7.44)$$

7.3 Maxwell's Equations

So far we have 4 equations:

$$\begin{aligned} (i) \quad \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} && \text{(Gauss's law),} \\ (ii) \quad \nabla \cdot \mathbf{B} &= 0 && \text{(No name),} \\ (iii) \quad \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} && \text{(Faraday's law),} \\ (iv) \quad \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} && \text{(Ampère's law).} \end{aligned}$$

But there is a problem with Ampère's law, since it is only for magnetostatics. Maxwell fixed this problem and the (iv) equation becomes:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (7.46)$$

Griffiths write the Maxwell equations like this:

$$\left. \begin{aligned} (i) \quad \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0}, & (iii) \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= \mathbf{0}, \\ (ii) \quad \nabla \cdot \mathbf{B} &= 0, & (iv) \quad \nabla \times \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} &= \mu_0 \mathbf{J}. \end{aligned} \right\} \quad (7.52)$$

Maxwell's Equations in Matter

If the polarization of a material changes it induces a current:

$$\mathbf{J}_p = \frac{\partial \mathbf{P}}{\partial t}, \quad (7.58)$$

which means that the total current density is:

$$\mathbf{J} = \mathbf{J}_f + \mathbf{J}_b + \mathbf{J}_p = \mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t}. \quad (7.60)$$

With this we can write Maxwell's equations for free charge and current:

$$\left. \begin{aligned} (i) \quad \nabla \cdot \mathbf{D} &= \rho_f, & (iii) \quad \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ (ii) \quad \nabla \cdot \mathbf{B} &= 0, & (iv) \quad \nabla \times \mathbf{H} &= \mathbf{J}_f + \frac{\partial \mathbf{H}}{\partial t}, \end{aligned} \right\} \quad (7.65)$$

where

$$\begin{aligned} \mathbf{D} &= \varepsilon \mathbf{E} \quad \text{and} \quad \mathbf{H} = \frac{1}{\mu} \mathbf{B}, \\ \varepsilon &= \varepsilon_0(1 + \chi_e) \quad \text{and} \quad \mu = \mu_0(1 + \chi_m). \end{aligned} \quad (7.67)$$

8 Conservation Laws

8.1 Charge and energy

Because of local charge conservation, charge must flow through a surface to leave a system:

$$\frac{dQ}{dt} = - \oint_S \mathbf{J} \cdot d\mathbf{a}, \quad (8.2)$$

which leads to the continuity equation:

$$\frac{d\rho}{dt} = -\nabla \cdot \mathbf{J}. \quad (8.4)$$

From the work required to assemble a static charge distribution (2.45), and the work required to get current to move against the back emf (7.44), we get the total stored energy in electromagnetic fields, per unit volume:

$$u = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right). \quad (8.5)$$

The rate of work done on all charges in a volume is:

$$\frac{dW}{dt} = \int \mathbf{E} \cdot \mathbf{J} d\tau, \quad (8.6)$$

which can be rewritten using Ampère-Maxwell law:

$$\frac{dW}{dt} = -\frac{d}{dt} \int_V \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) d\tau - \frac{1}{\mu_0} \oint_S (\mathbf{E} \times \mathbf{B}) \cdot d\mathbf{a}. \quad (8.9)$$

This is Poynting's theorem. The first integral is the total energy stored in the electromagnetic field, the second integral is the energy leaving the volume V . The energy per unit time, per unit area, transported by the fields is called the Poynting vector:

$$\mathbf{S} = \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}). \quad (8.10)$$

Now we can rewrite (8.9) again:

$$\frac{dW}{dt} = -\frac{d}{dt} \int_V u d\tau - \oint_S \mathbf{S} \cdot d\mathbf{a}. \quad (8.11)$$

This leads to a general continuity equation for energy in electromagnetic fields:

$$\frac{\partial u}{\partial t} = -\nabla \cdot \mathbf{S}. \quad (8.12)$$

8.2 Momentum

The force per unit volume on charges is:

$$\mathbf{f} = \rho \mathbf{E} + \mathbf{J} \times \mathbf{B}, \quad (8.14)$$

$$\begin{aligned} &= \epsilon_0 [(\nabla \cdot \mathbf{E})\mathbf{E} + (\mathbf{E} \cdot \nabla)\mathbf{E}] + \frac{1}{\mu_0} [(\nabla \cdot \mathbf{B})\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{B}] \\ &\quad - \frac{1}{2} \nabla \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) - \epsilon_0 \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}). \end{aligned} \quad (8.16)$$

which can be rewritten using Maxwell stress tensor:

$$T_{ij} = \epsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right), \quad (8.17)$$

9 Appendix

Tables

Table 4.2. Dielectric constants (unless otherwise specified, values given are for 1 atm, 20°C).
Data from *Handbook of Chemistry and Physics*, 91st ed., CRC Press, Boca Raton (2010).

Material	Dielectric constant	Material	Dielectric constant
Vacuum	1	Benzene	2.28
Helium	1.000065	Diamond	5.7–5.9
Neon	1.00013	Sodium chloride	5.9
Hydrogen (H ₂)	1.000254	Silicon	11.7
Argon	1.000517	Methanol	33.0
Air (dry)	1.000536	Water	80.1
Nitrogen (N ₂)	1.000548	Ice (−30°C)	104
Water vapor (100°C)	1.00589	KTaNbO ₃ (0°C)	34,000

Table 6.1. Magnetic susceptibilities (unless otherwise specified, values are for 1 atm, 20°C).
Data from *Handbook of Chemistry and Physics*, 91st ed., CRC Press, Boca Raton (2010)
and other references.

Material	Susceptibility	Material	Susceptibility
<i>Diamagnetic:</i>		<i>Paramagnetic:</i>	
Bismuth	-1.7×10^{-4}	Oxygen (O ₂)	1.7×10^{-6}
Gold	-3.4×10^{-5}	Sodium	8.5×10^{-6}
Silver	-2.4×10^{-5}	Aluminum	2.2×10^{-5}
Copper	-9.7×10^{-6}	Tungsten	7.0×10^{-5}
Water	-9.0×10^{-6}	Platinum	2.7×10^{-4}
Carbon dioxide	-1.1×10^{-8}	Oxygen (liquid) (−200°C)	3.9×10^{-3}
Hydrogen (H ₂)	-2.1×10^{-9}	Gadolinium	4.8×10^{-1}

Table 7.1. Resistivities, in ohm meters (all values are for 1 atm, 20°C). Data from *Handbook of Chemistry and Physics*, 91st ed., CRC Press, Boca Raton (2010) and other references.

Material	Resistivity	Material	Resistivity
<i>Conductors:</i>		<i>Semiconductors:</i>	
Silver	1.59×10^{-8}	Sea water	0.2
Copper	1.68×10^{-8}	Germanium	0.46
Gold	2.21×10^{-8}	Diamond	2.7
Aluminum	2.65×10^{-8}	Silicon	2500
Iron	9.61×10^{-8}	<i>Insulators:</i>	
Mercury	9.61×10^{-7}	Water (pure)	8.3×10^3
Nichrome	1.08×10^{-6}	Glass	$10^9\text{--}10^{14}$
Manganese	1.44×10^{-6}	Rubber	$10^{13}\text{--}10^{15}$
Graphite	1.6×10^{-5}	Teflon	$10^{22}\text{--}10^{24}$